



 slington college
(इस्लिङ्टन कलेज)

Module Code & Module Title
Level 6 – Artificial Intelligence

Coursework 02
Semester
2024/25 Autumn

Student Name: Prapti Sangraula

London Met ID: 22085565

College ID: NP01CP4S230032

Assignment Due Date: Wednesday, January 22, 2025

Assignment Submission Date: Tuesday, January 21, 2025

Submitted To: Mr. Alish KC, Mr. Dipeshor Silwal

Word Count (Where Required): 6177

I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

15% Overall Similarity

 Filters

Match Groups	Sources
--------------	---------

71 matches found with Turnitin's database

Show Help 

	61 Not Cited or Quoted	13%
	8 Missing Quotations	2%
	0 Missing Citation	0%
	2 Cited and Quoted	1%

Table of Contents

1. Introduction	1
2. Background	3
3. Solution	16
4. Conclusion	39
5. References.....	40

Table Of Figures

Figure 1: Reasons for dropping out of college (Winograd, 2024)	1
Figure 2: Dataset description	7
Figure 3: Dataset description 2	8
Figure 4: Dataset description 3	9
Figure 5: Formula of Accuracy	10
Figure 6: Formula of Recall	10
Figure 7: Formula of Precision	11
Figure 8: Formula of F1 score	11
Figure 9: Block Diagram.....	15
Figure 10: Formula of Euclidean distance	17
Figure 11: Flowchart of KNN	18
Figure 12: Formula of Bayes Theorem.....	19
Figure 13: Flowchart of Naive Bayes.....	21
Figure 14: Decision Tree (IBM, 2024)	22
Figure 15: Random Forest (Riebesell, 2024)	23
Figure 16: Flowchart of Random Forest.....	25
Figure 17: Importing Libraries and loading the data	27
Figure 18: Cleaning the data	27
Figure 19: Handling missing values of the data.....	28
Figure 20: Checking for duplicate values in the data.....	28
Figure 21: No missing or duplicate values were found	28
Figure 22: Visualizing distribution of the target variable	29
Figure 23: Result of visualization of target variable	29
Figure 24: Creation of a correlation matrix	30
Figure 25: Visualization results of correlation matrix	31
Figure 26: Defining the features variables and target variables	32
Figure 27: Splitting the data	32
Figure 28: Standardizing the data	33
Figure 29: Initializing models.....	33
Figure 30: Visualizing important features of the data	34

Figure 31: Results of visualizing important features	34
Figure 32: Evaluating the models	35
Figure 33: Evaluation result of KNN	35
Figure 34: Evaluation Result of Naive Bayes	36
Figure 35: Evaluation result of random forest.....	37
Figure 36: Making predictions using test input	37
Figure 37: Result of the prediction	38

Table Of Tables

Table 1: Evaluation of Study 1	12
Table 2: Evaluation of Study 2	13
Table 3: Evaluation of Study 3	14
Table 4: Comparison of results	14
Table 5: Why algorithms were chosen	26

1. Introduction

The topic of this project is to predict the academic outcomes of students by using machine learning algorithms. There has been a big rise in decision making done through the usage of data. This is due to the usage of Artificial Learning (AI). Artificial Intelligence has revolutionised numerous domains. This includes the domain of education. For the domain of education, large data sets can be analysed in order to get insights, predict future trends and make strategic decisions. The aim of this project is to successfully predict student dropouts using previous data. This will help the educational institutions to provide support to students who are at a risk of dropping out.

This project explores the problem domain of Student dropout. Student dropout is a huge socioeconomic issue around the world. There can be various reasons as to why a student may choose to dropout and quit studies. These reasons may include financial issues, mental health issues or motivation or life changes. Issues students can have include cost of tuition, cost of additional expenses and cost of living. Issues that students may face related to their mental wellbeing/ mental health include Depression, Anxiety and Substance Abuse, All of these reasons have the ability to lead a student to dropping out. An article from mission graduate conducted a survey which explored the reasons behind dropout rates. The result of the survey is given below:

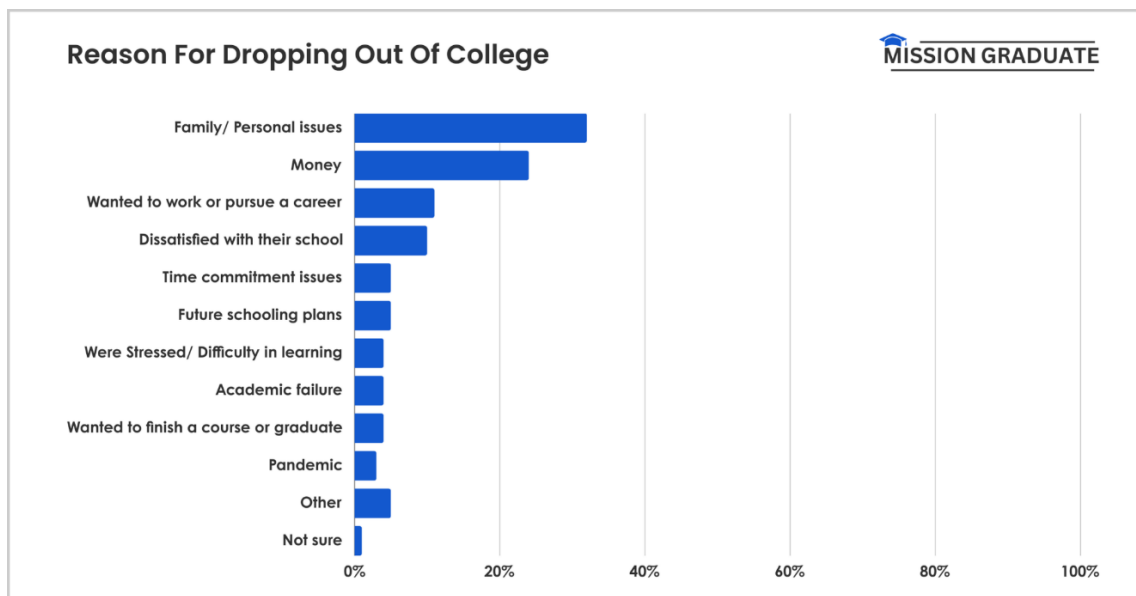


Figure 1: Reasons for dropping out of college (Winograd, 2024)

This survey showed that 32 percent of the students dropped out of the college due to personal or family issues, 24 percent of the students dropped out due to financial issues, 11 percent of the students dropped out due to wanting to pursue another career, 10 percent of the students dropped out because they were dissatisfied with the institution and so on.

Ensuring that students finish their education is known to be very important for their futures. Student dropout has the potential to negatively impact the lives of the students and the institutions where the students are enrolled. For students, it can limit their future employment opportunities and wellbeing. For educational institutions, high dropout rates can appear poorly on their records and affect their reputation. On a societal level, high dropout rates can hinder economic growth.

Because this is such a big challenge it is no surprise that research regarding student dropout has been going on for several decades. Early research on this topic dates as far as the 1970s (Tinto, 1975). As time has passed, this research has expanded and the use of modern technologies, computing power and data collection techniques has further enhanced the ability to study the problem more accurately.

The primary aim of this project is to develop and assess machine learning models to forecast student dropout based on historical data. This is to help educational institutions to take early measures.

The objectives of this project are outlined below:

- To compare three machine learning algorithms for accuracy and effectiveness based on various relevant metrics.
- To evaluate the effectiveness of all of the three models using performance metrics.
- To successfully identify students at risk for dropping out using machine learning models.

2. Background

This project covers the field of predictive analysis. Predictive analysis is the process where data is used in order to answer questions, to make strategies and in order to make impactful decisions. Predictive analysis is often associated with the field of big data and computer science. Big data is extremely large sets of data which is too complicated to perform analysis on using traditional data processing methods. Projects related to big data demand intense resources for processing and storage. A lot of speed, power and flexibility is required in order to quickly access huge amounts of data. This is all provided by modern computing systems. Algorithms play a huge role in big data projects. They provide the methods to access, analyse and derive valuable insights from large and complicated datasets. Without using algorithms, it would be extremely hard and overwhelming to use the data. Algorithms find specific patterns to extract insights from data and forecast future events (IBM, 2024). Regardless of the field, predictive analysis proves useful in providing insights to make decisions (Cote, 2024). Advancements in machine learning have helped expand predictive analytics capabilities. Popular predictive analysis models include classification and clustering. The former falls under supervised learning and the latter falls under unsupervised learning. (IBM, 2024).

Supervised Learning

This type of learning makes use of datasets that have labels in order to train algorithms for accurate data classification.

Classification models

Classification models belong to the field of supervised learning. Classification refers to grouping data into certain already defined categories on the basis of their specific attributes. An example of a use case of this model would be using it to classify prospects or customers into groups for segmentation. Another example of a use case can be to simply use it to answer yes or no questions for things like fraud detection (IBM, 2024).

Unsupervised Learning

This type of learning makes use of datasets which do not have labels. Unsupervised learning algorithms have no previous knowledge regarding the data and are tasked with finding patterns and relationships between the data (geeksforgeeks, 2024).

Clustering models

Clustering models fall under the branch of unsupervised machine algorithms. Clustering involves grouping data based on similarities and dissimilarities into clusters (Simplilearn, 2024). An example of a use case of this model could be to separate customers into similar groups for marketing effectively to each group.

The advantages of using AI in this domain are:

- Early Intervention to provide required support for students who are at risk
- Make decisions which are driven by the data.
- AI models are able to easily handle large datasets.

The possible drawbacks that may arise due to the usage of AI in this domain are:

- Data related to students is sensitive and can cause some data privacy concerns if it is not handled carefully.
- The accuracy of the model might be affected by data imbalance caused due to some outcomes having fewer samples.

In this project, the focus is on applying predictive analysis techniques to predict student dropout. To achieve this, three supervised learning algorithms are going to be used. All three algorithms that are going to be used fall under Supervised learning under the subcategory of Classification Models. An introduction to the models is provided below and they will be further explored in the solution section.

K-Nearest Neighbours (KNN): This is a supervised learning algorithm which makes classifications or predictions based on the closeness of the data points. KNN is appropriate for both classification and regression tasks, but it is primarily utilized as a classification algorithm.

Naïve Bayes: This is a supervised learning algorithm that functions according to the Bayes' Theorem. Bayes' Theorem is employed to determine conditional probability (Chauhan, 2022). Conditional probability refers to the chance of an event happening based on a previous result in comparable conditions (Hayes, 2024).

Random Forest: Random Forest: A random forest consists of multiple decision trees that collaborate and function collectively to yield a single result (geeksforgeeks, 2024). By using these models, the project aims to provide a way to predict student dropouts with a goal of improving educational outcomes.

- However, as with any model, there are limitations and areas for improvement. For this student dropout prediction model, these include:
- The accuracy of this model depends on availability of reliable and comprehensive student data. This might not always be realistic.
- Although Random Forest and KNN are generally efficient, they can be a bit intensive, particularly with large datasets, and may require runtime and memory optimization.
- The ensemble structure of Random Forest might make it more difficult to grasp, which may be an issue for stakeholders that want clear explanations of model predictions.

Dataset Information

The dataset which is used is a historical student dataset which comes from a higher educational institution. The dataset contains details of students participating in various courses. These programs include agronomy, design, social service and many more. The goal for the creation of the dataset was to help to reduce student dropout and failure by using machine learning technologies.

The dataset covers demographic variables such as Marital status and Nationality. Marital status may cause variations in financial stability and time constraints due to which dropout and graduation rates may differ between married and unmarried students. Different nationalities may face language barriers or cultural integration challenges which can make certain nationality groups experience higher dropout rates compared to others. Academic and socioeconomic variables such as previous qualifications, parental income are also covered by the dataset. Students with better achievements or prior degrees may

show lower dropout rates and higher graduation rates. And variables such as the occupation of the mother or the occupation of the father may show certain occupation groups experiencing higher dropout rates compared to others. The target variable is divided into dropout, graduate and enrolled which reflect outcomes of student enrolment. By using these features and analysing them, educational institutions can figure out what factors cause most of the influence in leading students to dropping out and they can tailor their decisions accordingly. The dataset includes 37 columns and 4424 instances. Each instance of the dataset represents a student. Short descriptions of all of the columns of this dataset are given starting the next page.

Column Name	Column Description
Marital Status	Marital status: 1 - Single, 2 - Married, 3 - Widower, 4 - Divorced, 5 - Facto Union, 6 - Legally Separated
Application Mode	Application mode with specific codes representing various types of admission
Application Order	Application order (between 0 - first choice; and 9 last choice)
Course	Course identifier with specific codes for various programs
Daytime/evening attendance	1 – daytime 0 – evening
Previous Qualification	Previous qualification level (e.g., Secondary education, Higher education)
Previous Qualification (grade)	Grade of previous qualification (between 0 and 200)
Nacionality	Nationality codes (e.g., 1 - Portuguese, 2 - German, etc.)
Mother's Qualification	Mother's education level (e.g., Secondary education, Higher education)
Father's Qualification	Father's education level (e.g., Secondary education, Higher education)
Mother's Occupation	Mother's occupation (coded with various professional categories)
Father's Occupation	Father's occupation (coded with various professional categories)
Admission Grade	Admission grade (between 0 and 200)
Displaced	1 - Yes, 0 - No (Whether the student is displaced)

Figure 2: Dataset description

Educational special needs	1 - Yes, 0 - No (Whether the student has special educational needs)
Debtor	1 - Yes, 0 - No (Whether the student owes fees)
Tuition fees up to date	1 - Yes, 0 - No (Whether tuition fees are up-to-date)
Gender	1 - Male, 0 - Female
Scholarship holder	1 - Yes, 0 - No (Whether the student is a scholarship holder)
Age at enrolment	Age of the student at enrolment
International	1 - Yes, 0 - No (Whether the student is an international student)
Curricular <u>units</u> 1 st semester (credited)	Number of curricular units credited in the 1st semester
Curricular <u>units</u> 1 st semester (enrolled)	Number of curricular units enrolled in the 1st semester
Curricular <u>units</u> 1 st semester (evaluations)	Number of evaluations to curricular units in the 1st semester
Curricular <u>units</u> 1 st semester (approved)	Number of curricular units approved in the 1st semester
Curricular <u>units</u> 1st semester (grade)	Grade average in the 1st semester (between 0 and 20)
Curricular <u>units</u> 1st semester (without evaluations)	Number of curricular units without evaluations in the 1st semester
Curricular <u>units</u> 2nd semester (credited)	Number of curricular units credited in the 2nd semester
Curricular <u>units</u> 2nd semester (enrolled)	Number of curricular units enrolled in the 2nd semester

Figure 3: Dataset description 2

Curricular <u>units</u> 2nd semester (evaluations)	Number of evaluations to curricular units in the 2nd semester
Curricular <u>units</u> 2nd semester (approved)	Number of curricular units approved in the 2nd semester
Curricular <u>units</u> 2nd semester (grade)	Grade average in the 2nd semester (between 0 and 20)
Curricular <u>units</u> 2nd semester (without evaluations)	Number of curricular units without evaluations in the 1st semester
Unemployment rate	Unemployment rate (%)
Inflation rate	Inflation rate (%)
GDP	GDP
Target	Target. The problem is formulated as a <u>three-category</u> classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course

Figure 4: Dataset description 3

Over the past twenty years, with the increased accessibility of digital data, analysing the reasons behind student dropouts has become simpler. Institutions of higher learning collect extensive data on their students, which presents a substantial opportunity to produce information, derive insights, and conduct monitoring (Realinho, et al., 2022). This has been made achievable through using machine learning. Researchers have examined machine learning techniques which are traditional like Naïve Bayes, Logistic Regression, Decision Trees (DT), K-Nearest Neighbours, Random Forest, and Support Vector Machines (SVM) to create predictive models for the purpose of forecasting student dropout. Other researchers have also taken leverage of neural network methods such as CNN (Convolutional Neural Network) to predict student dropout (Chi, et al., 2023). Several notable studies and projects have focused on the area of Student Dropout Prediction. All of these make use of certain evaluation metrics to assess their models. The metrics include recall, precision, F1 score, AUC and Accuracy. To understand the results that are given by the metrics, understanding the concepts of True Positives, False Positives, True Negatives and False negatives is essential.

True Positive: This is when the prediction had a positive outcome and the result also had a positive outcome.

True Negative: This is when the prediction had a negative outcome and the result also had a negative outcome.

False Positive: This is when the prediction had a positive outcome but the result did not have a positive outcome and was negative.

False Negative: This is when the prediction had a negative outcome but the actual result did not have a positive outcome and was negative.

Now, the metrics used are explained below:

Accuracy: Accuracy is a machine learning evaluation metric that tells us how often a model makes the right prediction compared to the total prediction that the model makes. Accuracy is measured by the formula given below:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total Number of predictions}}$$

Figure 5: Formula of Accuracy

Recall: Recall is a machine learning evaluation metric which determines the proportion of actual positives accurately identified as positives by the model. Recall is calculated using the formula provided below:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False negatives}}$$

Figure 6: Formula of Recall

Precision: Precision is a machine learning evaluation metric that calculates the proportion of true positive predictions compared to the total number of positive predictions made by the model. Precision is measured by the formula below:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False positives}}$$

Figure 7: Formula of Precision

F1 score: This is a machine learning evaluation metric which accounts for both precision and recall. The F1 score can be calculated using the following formula:

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 8: Formula of F1 score

AUC: AUC shows the likelihood that when a positive and negative example are randomly picked, the model will rank the positive example higher than the negative one (Google Developer Program, 2024)

The notable studies that apply these metrics to assess models for student dropout prediction are:

Study 1: “Early Dropout Prediction in Online Learning of University using Machine Learning” (Park & Yoo, 2021)

Description: This study focuses on predicting dropout rates among students of online universities in Korea. For more than a decade, data was collected in the learning management system (LMS) of Korea's online colleges which is utilized in this (Park & Yoo, 2021). The machine learning algorithms that were used for this study are Deep Neural Network, Random Forest and Support Vector Machines.

The performance of the models was judged using accuracy, recall, prevision, AUC and F1score. The following are the results of the evaluation metrics:

Study	Model	Accuracy	Recall	Precision	F1 Score	AUC
Early Dropout Prediction in Online Learning of University using Machine Learning	Decision Tree	0.91	0.70	0.66	0.68	0.82
	Random Forest	0.96	0.76	0.96	0.84	0.95
	SVM	0.81	0.55	0.31	0.40	0.69
	DNN	0.85	0.39	0.41	0.37	0.77

Table 1: Evaluation of Study 1

From these results, it is evident that the algorithm which has the best values in most of the evaluation metrics is Random Forest. At the end of the study, the conclusion was made that Random Forest was the best for this prediction.

Study 2: “Towards Predicting Student’s Dropout in University Courses Using Different Machine Learning Techniques” (Kabathova & Drlik, 2021)

Description: This study investigated dropout prediction among students of LMS Moodle’s e-learning course. The machine learning algorithms that were used for this study are Naïve Bayes, Random Forest, Neural Network, logistic regression, Support vector machine and Decision Tree.

The performance of the models was judged using accuracy, precision, recall and F1 score. The following are the results of the evaluation metrics:

Study	Model	Accuracy	Recall	Precision	F1 Score
Towards Predicting Student's Dropout in University Courses Using Different Machine Learning Techniques	Naïve Bayes	0.77	0.72	0.93	0.82
	Random Forest	0.93	0.96	0.86	0.91
	Neural Network	0.88	0.89	0.86	0.88
	Logistic Regression	0.93	0.98	0.79	0.87
	Support Vector Machine (SVM)	0.92	0.96	0.79	0.88
	Decision Tree (DT)	0.90	0.98	0.71	0.85

Table 2: Evaluation of Study 2

The three algorithms that evidently perform the best are Logistic Regression, Random Forest and Support vector machines. Among them, Random Forest shows high recall and precision. Logistic Regression achieved the highest recall but has less precision. Support Vector Machines also performs well but has low precision. Hence, at the end of the study, Random Forest was the better model for this prediction as well.

Study 3: “Analysis and Prediction of MOOC Learners’ Dropout Behaviour (Chi, et al., 2023)

Description: This study analyses the data from a Massive Open Online Course (MOOC) to know what factors cause learner’s student dropout behaviour. It employs Random Forest, Logical Regression and K Nearest Neighbour (KNN) for the dropout prediction.

The performance of the models was judged using accuracy, precision, recall, AUC and F1 score. The following are the results of the evaluation metrics:

Study	Model	Accuracy	Recall	Precision	F1 Score	AUC
Analysis and Prediction of MOOC Learners' Dropout Behaviour	Random Forest	0.92	0.99	0.93	0.95	0.93
	Logistic Regression	0.91	0.98	0.93	0.95	0.91
	KNN	0.92	0.98	0.93	0.95	0.93

Table 3: Evaluation of Study 3

From these results, evidently, Random Forest achieved the best overall performance across metrics. It had almost a perfect recall of 0.99. KNN and Logistic were also good however Random Forest stood out from the three. At the end of the study, it was concluded that the prediction effect of Random forests was found to be better than that of KNN and logical regression model.

Comparison of Results

Study	Best performing Model	Accuracy	Recall	Precision	F1 score	AUC
Study 1	Random Forest	0.96	0.76	0.96	0.84	0.95
Study 2	Random Forest	0.93	0.96	0.86	0.91	Not Given
Study 3	Random Forest	0.92	0.99	0.93	0.95	0.93

Table 4: Comparison of results

In all three studies, Random Forest has consistently demonstrated the highest effectiveness in forecasting student dropout. Alternative algorithms like KNN, Logistic Regression, and Support Vector Machines (SVM) also yield good results, but Random Forest has continually excelled beyond them.

For this system, a block diagram has also been created in order to provide clarify and to simplify the complex workflows.

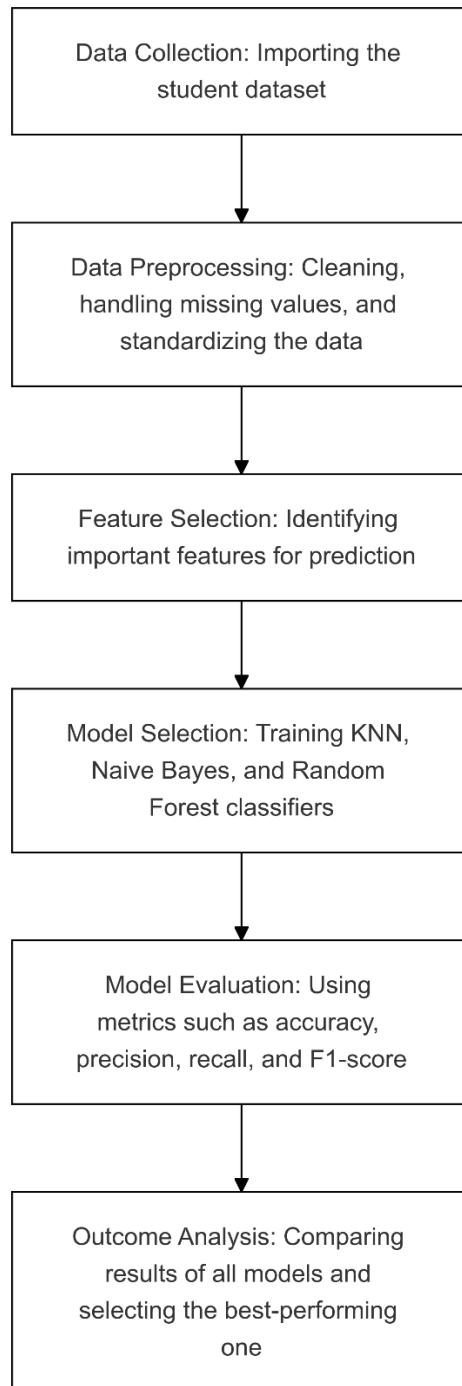


Figure 9: Block Diagram

3. Solution

The goal of this project is to combat high student dropout rates by building a model that makes predictions about which students might drop out. The predicting was done based on demographic factors such as Marital status and Nationality, academic and socioeconomic factors such as previous qualifications, parental income as well as other factors provided by the dataset. To achieve this, three popular basic algorithms were used. They are: Naïve Bayes, K-Nearest Neighbours (KNN) and Random Forest.

The algorithms assisted in analysing student data and identifying trends that can forecast whether a student will remain enrolled or dropout. Each of these algorithms worked differently and produced different results. In the end, by comparing the algorithms through various assessment metrics, the most suitable algorithm for this particular problem was identified. This approach to the solution provides a final reliable model for predicting student dropouts.

Before diving into the algorithms to be used, it is essential to understand these terms:

Overfitting: Overfitting is when a machine learning model performs good on training data but it does not perform well on any new data (aws, 2024).

Underfitting: Underfitting is a behaviour in which model is too simple to grasp the complexities of the data (geeksforgeeks, 2024) .

The algorithms used for prediction model are as follows:

K-Nearest Neighbours (KNN): This is a supervised learning algorithm which makes classifications or predictions based on the closeness of the data points. KNN is appropriate for both classification and regression tasks, but it is primarily utilized as a classification algorithm. When a new data point arrives, KNN determines the distance of this point from all existing ones utilizing the Euclidean distance equation:

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + \dots + (z_2 - z_1)^2]}$$

Where:

$x_1, x_2, y_1, y_2, \dots, z_1, z_2$ are the coordinates of two points

d is the Euclidean distance between the two points.

Figure 10: Formula of Euclidean distance

KNN identifies the k nearest points known as "neighbours" to the data. " k " signifies the count of neighbours. When selecting the value of " k ," the subsequent factors need to be taken into account:

- A low k value may be excessively particular, resulting in overfitting.
- A high k value may be overly generalized, resulting in underfitting.

To determine the optimal k value, the following approaches can be utilized:

- Cross Validation: Evaluating various k values to identify the most effective one.
- Elbow Method: Graphing the error rates for different k values. When k is increased, the error rate first drops but then levels off or begins to rise slightly. The optimal value of k is regarded as the "elbow" point of this curve. This is the point at which the error rate is minimal, and raising k further yields no advantages.
- The data point is given the label that is most frequent among these members.

Pseudocode of KNN:

SET UP the KNN classifier with a specified value for k .

FIT the KNN model to the training dataset.

ITERATE over each test example

CALCULATE distance using the Euclidean distance formula

IDENTIFY the k nearest neighbours based on distance

CONDUCT majority voting in order to identify the predicted class.

EVALUATE the performance of the model

OUTPUT the evaluation results

The flowchart of KNN:

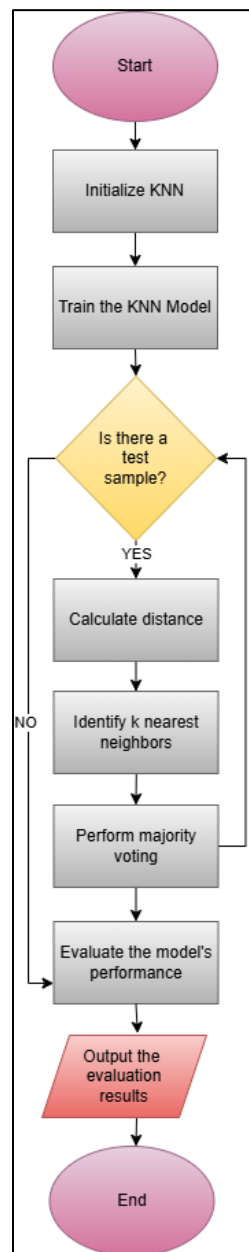


Figure 11: Flowchart of KNN

Naïve Bayes: This is a supervised learning algorithm that functions according to the Bayes' Theorem. Bayes' Theorem is employed to determine conditional probability (Chauhan, 2022). Conditional probability refers to the chance of an event happening based on a previous result in comparable conditions (Hayes, 2024).

Formula for Bayes Theorem:

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

Where:

$P(Y|X)$ represents the probability of Y (the label or outcome) given we have X (the input features)

$P(X|Y)$ represents the probability of observing the features (X) if we already know the outcome (Y)

$P(Y)$ represents the overall likelihood of Y occurring

$P(X)$ represents the overall likelihood of observed features X across all classes.

Figure 12: Formula of Bayes Theorem

The Naïve Bayes classifier is termed "Naïve" Bayes due to the assumptions it relies on. The Naïve Bayes classifier operates under two assumptions. One is that every feature contributes independently to the result. Another point is that all attributes contribute equally to the result (Chauhan, 2022). The varieties of Naïve Bayes classifier include:

Gaussian Naïve Bayes: This operates on numerical (continuous) data and presumes that numerical features adhere to a Gaussian (normal) distribution

Multinomial Naïve Bayes: This is suited for discrete data, used in applications such as text classification, and assumes that features denote the frequency of occurrences.

Bernoulli Naïve Bayes: This operates with binary features and presumes that every feature is binary (either 0 or 1).

Due to the naive assumption held by the Naïve Bayes classifier, this algorithm is extremely quick and effective. This is due to its need for less time for training and

predicting compared to many complex models. Nevertheless, this classifier presumes that every feature operates independently, which is often not the case for numerous datasets.

Pseudocode of Naïve Bayes:

SET UP the Naïve Bayes' classifier

FIT the model to the training dataset.

ITERATE over each class C in the target variable:

CALCULATE $P(C)$

ITERATE over every feature X

CALCULATE $P(X|C)$ using the Gaussian formula

PREDICT labels for the test data

ITERATE over each test example

CALCULATE $P(C|X) = P(X|C) * P(C)$

ASSIGN the class with the highest $P(C|X)$ as the predicted label.

EVALUATE the model's performance

OUTPUT the evaluation results

The flowchart for Naïve Bayes:

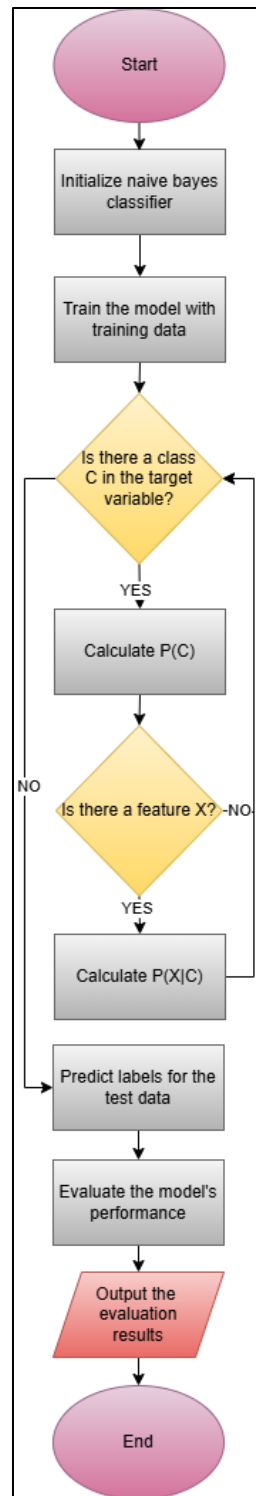


Figure 13: Flowchart of Naive Bayes

Random Forest: Random Forest: A random forest consists of multiple decision trees that collaborate and function collectively to yield a single result (geeksforgeeks, 2024).

A Decision Tree is a nonparametric supervised model. It includes a root node, branches, internal nodes, and leaf nodes (IBM, 2024). It can be utilized for both classification tasks and regression tasks (IBM, 2024). A decision tree starts from a root node. From the root node emerges a sequence of nodes known as decision nodes. Every decision node signifies a division point. From the decision nodes, leaf nodes emerge. The leaf nodes of the decision tree indicate the potential responses. Below is a diagram that illustrates a standard decision tree:

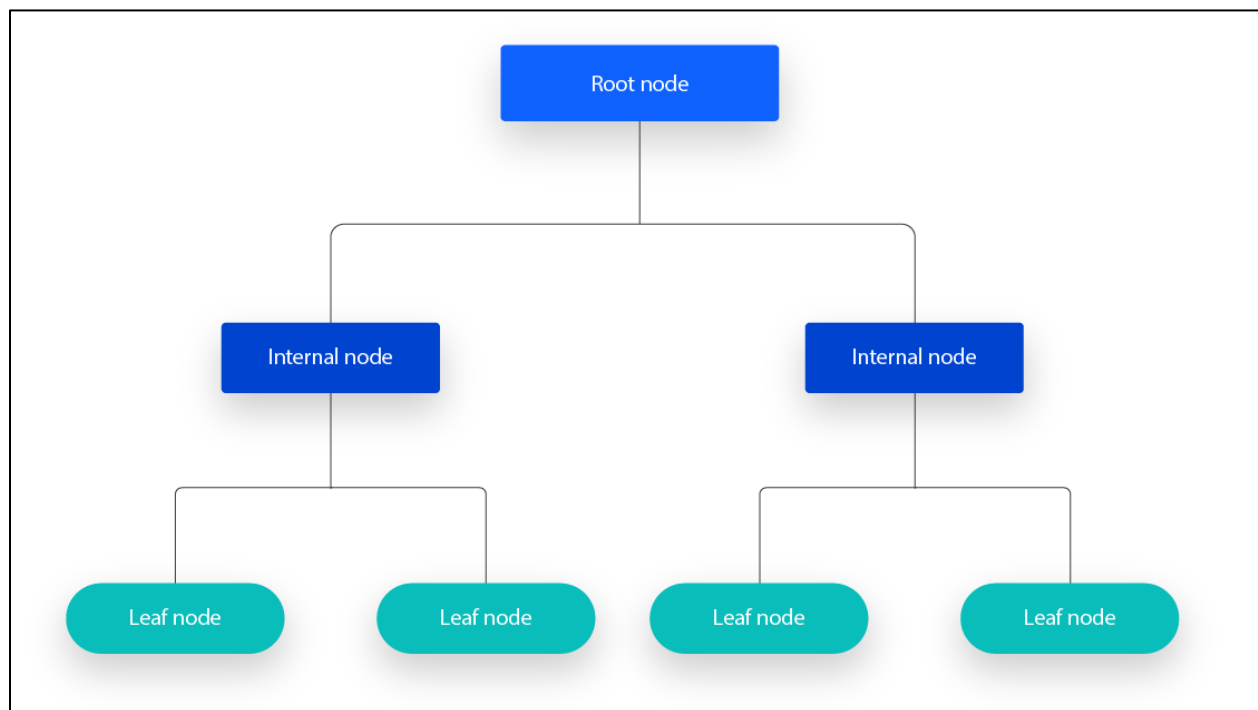


Figure 14: Decision Tree (IBM, 2024)

Random Forest falls under ensemble learning models. Ensemble learning consists of models, either of the same type or different, which team up to enhance predictive performance. Random forest operates by creating multiple trees throughout the training stage. Every tree in the random forest is generated using a random sample of the data. Randomness introduces variability among the trees, helping to avoid overfitting. Diversity among individual trees also enhances overall prediction accuracy.

To make predictions, the algorithm combines the outcomes of all trees. The method of combining predictions varies between classification issues and regression issues. In classification tasks, the outcomes from all trees are aggregated through majority voting. The class that receives the highest number of votes ultimately becomes the final prediction. On the other hand, for regression problems, the results from all the trees are combined by averaging. This process leads to the bias of the trees being smoothed out and gives a more accurate result. In this project, random forest is used for classification. Hence, the algorithm will aggregate the results of all trees by voting. Here is a diagram which represents a random forest:

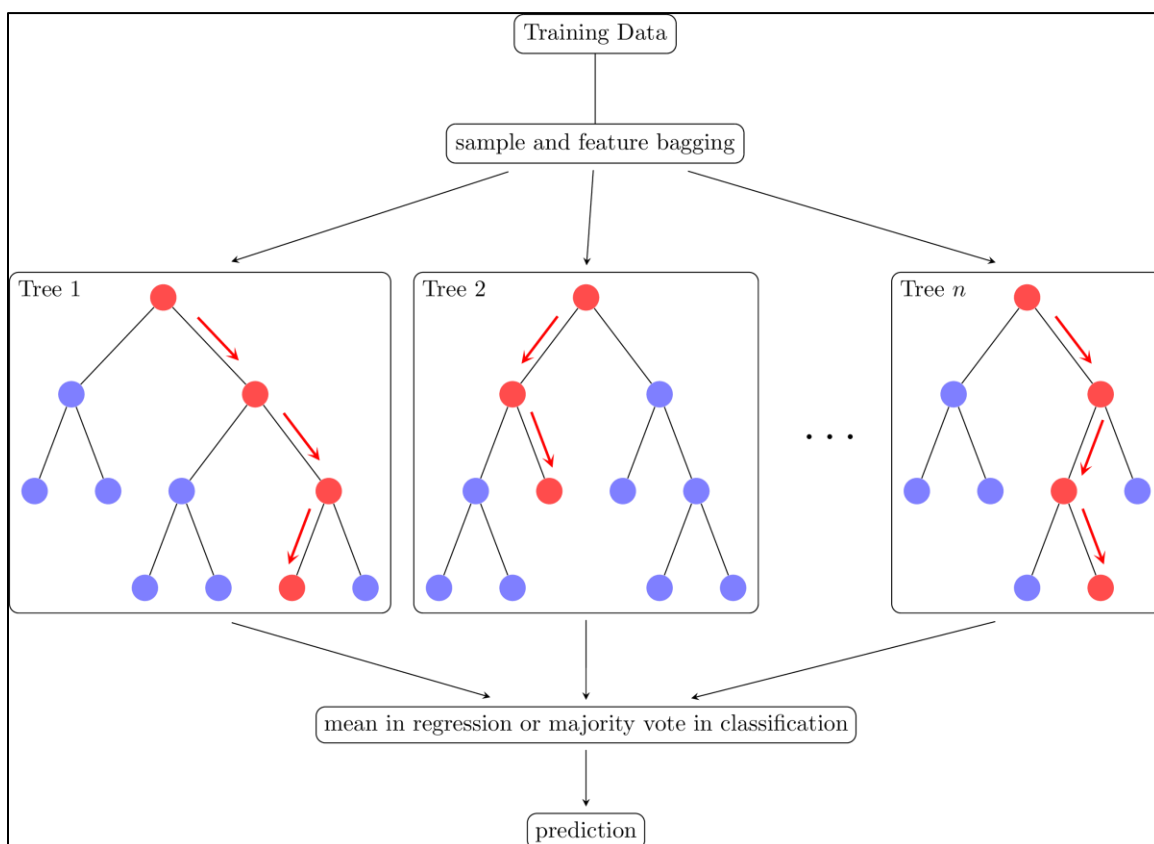


Figure 15: Random Forest (Riebesell, 2024)

Pseudocode of Random Forest:

SET UP the Random Forest classifier with T trees

FIT the Random Forest model to the training data

ITERATE over every tree in the forest

SELECT a random subset of training data

SELECT a random subset of features

BUILD the decision tree

PREDICT using all trees

ITERATE over each tree

OUTPUT the predicted class for the test sample

COMBINE predictions using majority voting

COUNT how many times each class appears in all the trees

SET the class that appears most frequently as the predicted label.

EVALUATE performance of the model

OUTPUT evaluation results

The Flowchart of Random Forest:

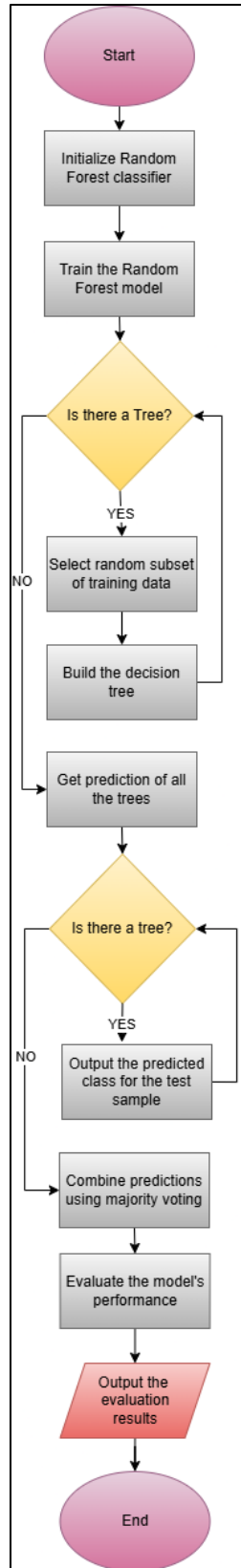


Figure 16: Flowchart of Random Forest

Why these algorithms were chosen:

Algorithm	Reason for choosing
KNN	KNN was selected because the dataset to be used for this project consists of numerical features that can effectively use proximity-based measures for classification.
Naïve Bayes	Naïve Bayes assumes feature independence which simplifies calculations and makes it computationally efficient.
Random Forest	Random forest combines the power of multiple decision trees which makes it a very robust choice.

*Table 5: Why algorithms were chosen***Platform and Language used:**

The development platform that was used in order to create this machine learning model was Jupyter Notebook. The language used was Python. The reason for choosing Python is because this provides a lot of libraries for machine learning, data handling, visualization. This makes the language the most ideal for this project.

The modules/libraries used

Pandas: This was used for loading tabular data and for manipulating it.

Numpy: Used for performing numerical operations and managing arrays.

Matplotlib: Used for visualizing data through basic plots.

Sklearn: This was used for machine learning models and for their evaluations.

Development Procedure

First, the necessary libraries to be used were imported and the dataset used was imported from a CSV file named data through using pandas. This step is absolutely crucial because it makes sure we have all of the correct and required libraries and dataset loaded correctly.

Importing the necessary libraries and loading the dataset:

```
1. Importing Libraries and Loading Data

# Importing necessary libraries
import pandas as pd
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

#Loading the file
data = pd.read_csv('data.csv', delimiter=';')
```

Figure 17: Importing Libraries and loading the data

The data cleaning involves making sure that the dataset is clean and consistent in order to analyse the data and to create good models. The data cleaning done involved:

Cleaning the names of the columns of the dataset in order to remove trailing spaces:

```
2. Data Cleaning and Preprocessing

# Cleaning the column names
data.columns = data.columns.str.strip() # Remove leading/trailing spaces
```

Figure 18: Cleaning the data

Handling missing values by removing the rows which contained those values:

```
# Checking for missing values and removing them if present
if data.isnull().values.any():
    print("\nThe dataset has missing values. These will be removed now.")
    data.dropna(inplace=True)
    print("Data after removing missing values:")
    print(data.head(10))
else:
    print("\nNo missing values detected.")
```

Figure 19: Handling missing values of the data

Checking and removing duplicate values:

```
# Checking for duplicate rows
if data.duplicated().any():
    print("\nDuplicate values were found. Removing them now.")
    # Display duplicate rows before removal
    print("Duplicate rows:")
    print(data[data.duplicated()].head(10))
    # Remove duplicates
    data.drop_duplicates(inplace=True)
    print("Duplicates have been removed.")
else:
    print("\nNo duplicate rows found.")
```

Figure 20: Checking for duplicate values in the data

No missing or duplicate values were found in the data.

```
No missing values detected.

No duplicate rows found.
```

Figure 21: No missing or duplicate values were found

Next, data visualization was performed to visualize the distribution of the target variable. This is important because it helps in seeing any class imbalances which are likely to impact model performance. In addition to visualizing the distribution of the target variable, a correlation matrix was also created in order to get insight on how strongly the relationships of features are with one another. First, target variable was visualized.

Visualizing Distribution of the target variable:

```
3. Data Visualization

import matplotlib.pyplot as plt

# Visualizing the distribution of the target variable
plt.figure(figsize=(8, 6))
data['Target'].value_counts().plot(kind='bar', color='pink')
plt.title('Distribution of Target Variable')
plt.xlabel('Target Categories')
plt.ylabel('Count')
plt.show()
```

Figure 22: Visualizing distribution of the target variable

Results:



Figure 23: Result of visualization of target variable

Findings

- The majority of the students in the dataset are labelled as “Graduate” which represents the largest group with more than 2000 entries.
- The “Dropout” category is the second largest group with approximately 1500 entries.
- The “Enrolled” category is the smallest group with fewer than 1000 entries.

Class imbalances of this sort may lead to overfitting of the dominant class and underperforming on the minority class. This means that the models may be biased towards the majority class which in this case is “Graduate”. Also, this can lead to poor performance in the “Dropout” and “Enrolled” classes which are the minority classes in this case. It may also lead to poor generalization on unseen data.

This can result in problems in the real-life applications of this prediction system as it might lead to missed opportunities for students who may be in need for attention. For example, a student who may be in risk of dropping out and who may need attention might be ignored because this system does not correctly predict the student’s outcomes. The solution to this has been explored in the [conclusion](#) section of this report.

Next, a correlation matrix was created.

Creating a correlation matrix:

```
# Correlation matrix
numeric_data = data.select_dtypes(include=[np.number])
corr_matrix = numeric_data.corr()
plt.figure(figsize=(15, 10))
plt.imshow(corr_matrix, cmap='coolwarm', interpolation='none')
plt.colorbar(label='Correlation Coefficient')
plt.xticks(range(len(corr_matrix)), corr_matrix.columns, rotation=90)
plt.yticks(range(len(corr_matrix)), corr_matrix.columns)
plt.title('Feature Correlation Matrix')
plt.tight_layout()
plt.savefig('correlation.png')
plt.show()
```

Figure 24: Creation of a correlation matrix

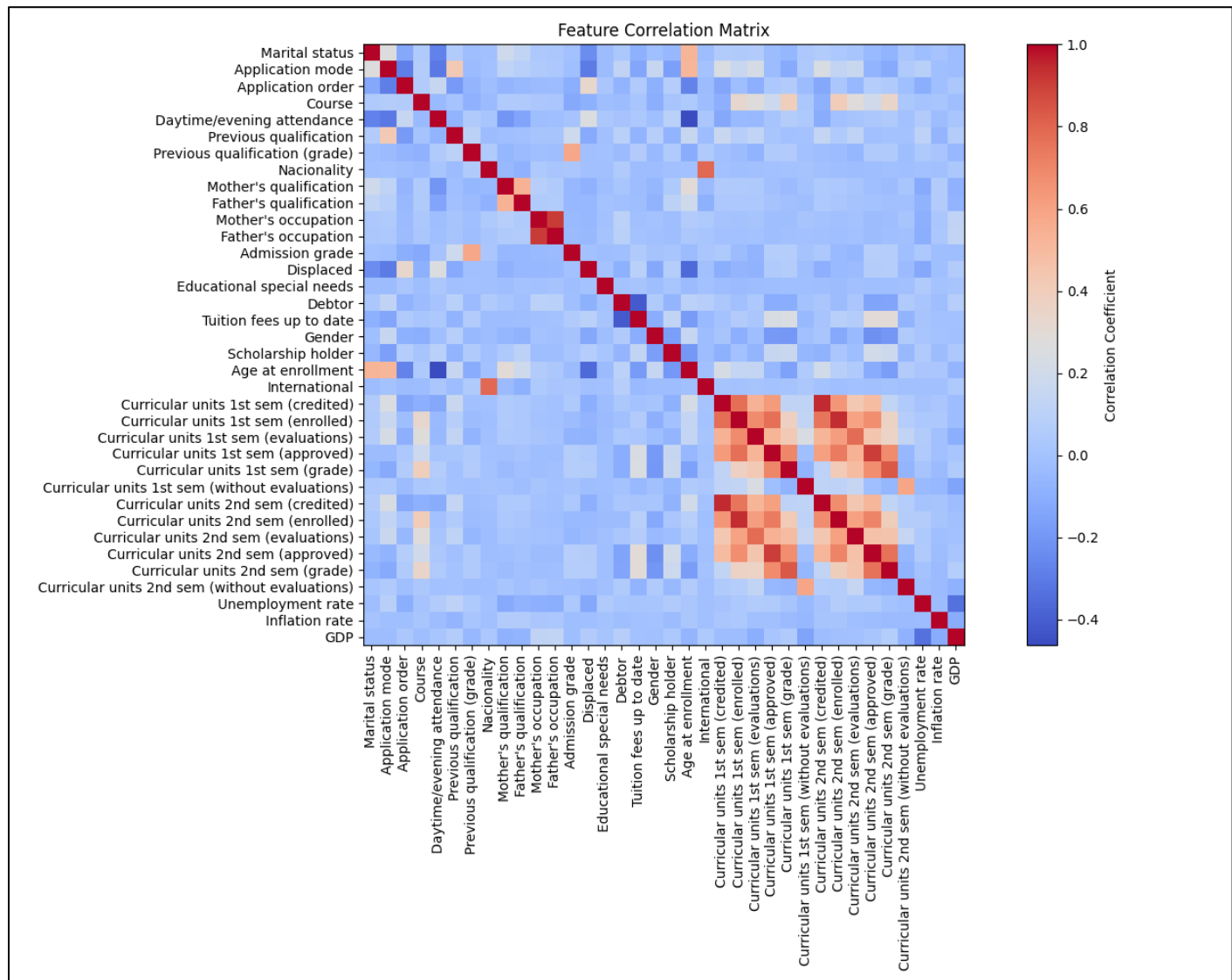
Results:

Figure 25: Visualization results of correlation matrix

The colour gradient used here is described shortly below:

- Red areas represent the stronger correlations which are closer to 1.
- Bluer areas represent the stronger negative correlations which are closer to -1.
- Lighter or neutral areas represent weaker or no correlations (close to 0).

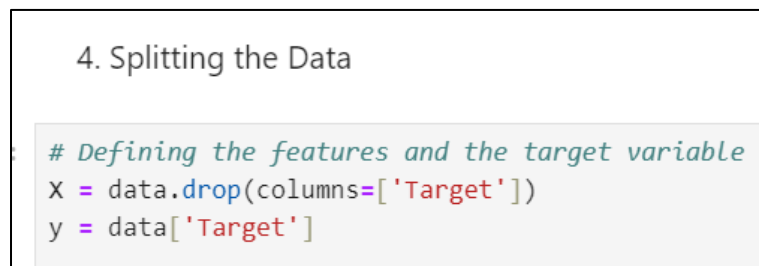
Findings:

- The diagonal cells are all red which denotes that it is a perfect correlation of 1.0. This is typical because each and every feature is perfectly correlated with itself.

- There are visibly strong squares near the very end of the correlation matrix. This indicates that the features near that area are very strongly linked. For example:
 - "1st sem (enrolled)", "1st sem (evaluations)", "1st sem (grade)", and "1st sem (approved)" appear to form a cluster. This suggests they are closely related.
 - This pattern repeats for the second semester as well. It is likely that these variables measure similar aspects of student performance or progression.
- Many of the cells are blue or light which indicates weak or even negative correlations.
 - For instance, demographic features like "Gender", "Marital status", or "Nationality" show limited correlation with academic features like "Grades" or "Evaluations."

Next, the features and target variable were defined to ensure that the data is structured correctly for model training.

Defining features and target variable:



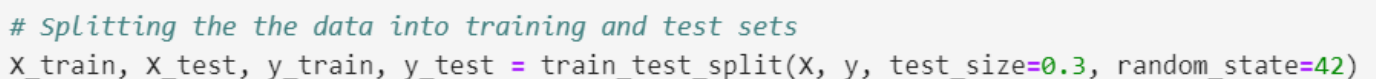
```
4. Splitting the Data

# Defining the features and the target variable
X = data.drop(columns=['Target'])
y = data['Target']
```

Figure 26: Defining the features variables and target variables

The data was then split into training and test sets. Splitting is done in order to train the model on one subset of data and then test it on another one.

Splitting data into training and testing sets:



```
# Splitting the the data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Figure 27: Splitting the data

The data was standardized to make sure that all features would equally contribute to the training. This is essential for sensitive machine learning algorithms such as KNN who are sensitive to the scale of the features.

Using Standard Scaler:

```
#Standardizing the data to ensure all features have equal weightage
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Figure 28: Standardizing the data

The machine learning models were then initialized and then trained using the .fit() method for each one of them.

Initializing the machine learning models:

5. Model Training

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.ensemble import RandomForestClassifier

# Initialize models
knn = KNeighborsClassifier(n_neighbors=5, weights='distance')
nb = GaussianNB()
rf = RandomForestClassifier(n_estimators=200, max_depth=20, random_state=42)

# Training models
knn.fit(X_train, y_train)
nb.fit(X_train, y_train)
rf.fit(X_train, y_train)
```

Figure 29: Initializing models

Visualizing important features:

Next, the important features were visualized using Random Forest. This helps identify the features that have the highest influence on the predictions. This helps identify the key factors that drive the behaviour of the model.

6. Visualizing feature importance

```
# Extracting feature importance from the Random Forest model
feature_importances = rf.feature_importances_

# Creating a dataframe for visualization
feature_names = X.columns
importance_df = pd.DataFrame({'Feature': feature_names, 'Importance': feature_importances})

# Sorting the features by importance
importance_df = importance_df.sort_values(by='Importance', ascending=False)

# Visualizing feature importance using a bar chart
plt.figure(figsize=(12, 6))
plt.bar(importance_df['Feature'], importance_df['Importance'], color='pink')
plt.title('Feature Importance from Random Forest')
plt.xlabel('Feature')
plt.ylabel('Importance Score')
plt.xticks(rotation=90)
plt.show()
```

Figure 30: Visualizing important features of the data

Results:

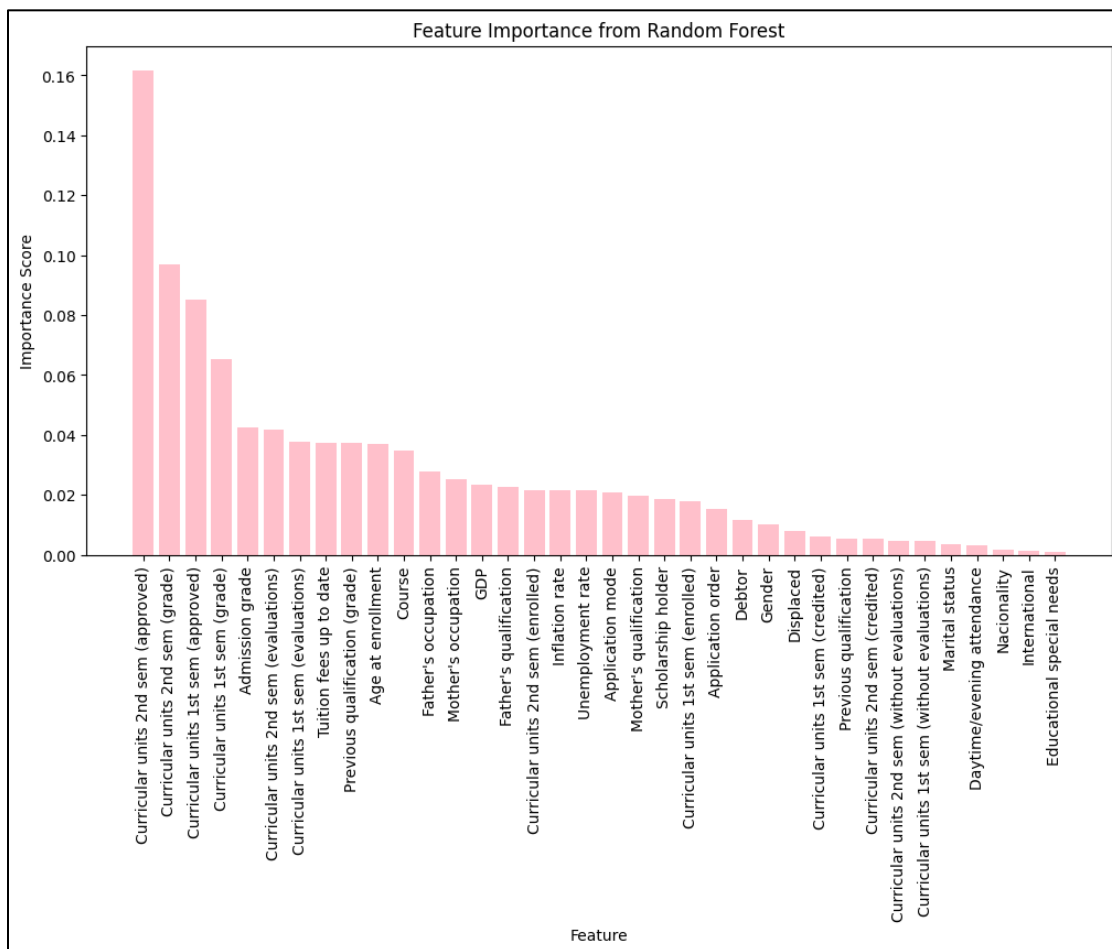


Figure 31: Results of visualizing important features

Next, each one of the models were evaluated to determine which model worked the best for this task.

Evaluating all three models:

```

7. Model Evaluation

from sklearn.metrics import accuracy_score, classification_report

# Evaluations
def evaluate_model(model, X_test, y_test):
    predictions = model.predict(X_test)
    print(classification_report(y_test, predictions))
    print(f"Accuracy: {accuracy_score(y_test, predictions)}\n")

print("KNN Evaluation:")
evaluate_model(knn, X_test, y_test)
print("Naive Bayes Evaluation:")
evaluate_model(nb, X_test, y_test)
print("Random Forest Evaluation:")
evaluate_model(rf, X_test, y_test)

```

Figure 32: Evaluating the models

Results of the evaluation:

These were the results of the evaluation:

KNN Evaluation:				
	precision	recall	f1-score	support
Dropout	0.83	0.66	0.73	441
Enrolled	0.39	0.28	0.33	245
Graduate	0.71	0.89	0.79	642
accuracy			0.70	1328
macro avg	0.65	0.61	0.62	1328
weighted avg	0.69	0.70	0.69	1328
Accuracy: 0.7010542168674698				

Figure 33: Evaluation result of KNN

For KNN, the accuracy result was 0.70 which is fairly good. But for each class there are differences in KNN's precision and recall.

The precision for dropout was 0.83 but recall was 0.66 which is lower comparatively. This shows that the model was good at recognizing dropout cases but it overlooked a lot of them. The graduate category had the highest recall. But the enrolled class had low precision and recall which shows that the model had a hard time in predicting this category.

The F1 score is the highest for graduate and the lowest for enrolled which shows that the model faced challenges in predicting the enrolled class.

Naive Bayes Evaluation:				
	precision	recall	f1-score	support
Dropout	0.79	0.70	0.74	441
Enrolled	0.37	0.28	0.32	245
Graduate	0.72	0.85	0.78	642
accuracy			0.69	1328
macro avg	0.63	0.61	0.61	1328
weighted avg	0.68	0.69	0.68	1328
Accuracy: 0.6927710843373494				

Figure 34: Evaluation Result of Naive Bayes

For Naïve Bayes, the model performed slightly worse than KNN and had a lower accuracy of 0.69. There were clear differences in the precision and recall across the classes just like KNN.

The precision for Dropout and recall for Graduate are higher and it shows that it performs well in these classes. However, it struggles with the Enrolled class just like KNN.

The F1 score is good for graduate and dropout but the significantly lower for enrolled which shows that the model faced challenges in predicting the enrolled class.

Random Forest Evaluation:				
	precision	recall	f1-score	support
Dropout	0.81	0.79	0.80	441
Enrolled	0.57	0.31	0.40	245
Graduate	0.78	0.93	0.85	642
accuracy			0.77	1328
macro avg	0.72	0.68	0.68	1328
weighted avg	0.75	0.77	0.75	1328
Accuracy: 0.7703313253012049				

Figure 35: Evaluation result of random forest

For Random Forest, it outperformed both KNN and Naïve Bayes with the best accuracy of 0.77.

The precision for Dropout and recall for Graduate are very high and it shows that it performs well in these classes. However, it still struggled with the Enrolled class with a recall of 0.31 and precision of 0.57.

The F1 score is very strong for graduate and dropout but the lower for enrolled which shows that there is still room for improvement in this category.

Making predictions using a test input:

The next step was to make predictions using a test input which was a string separated by commas. The predictions are displayed for each one of the models.

```

8. Providing Input to the models

# Providing Comma-separated values as input
input_string = "2,39,1,8014,0,1,100.0,1,37,38,9,9,141.5,0,0,0,1,0,0,45,0,0,6,9,5,12.333333333333334,0,0,6,6,6,13.0,0,

# Converting the string into a list of values by splitting by commas
student = [list(map(float, input_string.split(',')))]

# Making predictions using all three models
knn_prediction = knn.predict(student)[0]
nb_prediction = nb.predict(student)[0]
rf_prediction = rf.predict(student)[0]

# Displaying the predictions
print(f"KNN Prediction: {knn_prediction}")
print(f"Naive Bayes Prediction: {nb_prediction}")
print(f"Random Forest Prediction: {rf_prediction}")

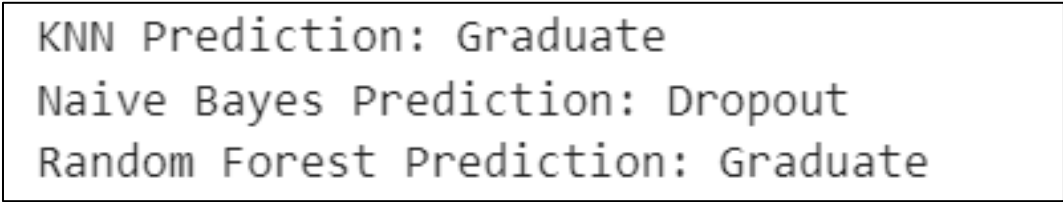
```

Figure 36: Making predictions using test input

This was the input string given:

"2,39,1,8014,0,1,100.0,1,37,38,9,9,141.5,0,0,0,1,0,0,45,0,0,6,9,5,12.333333333333333334,0,0,6,6,6,13.0,0,13.9,-0.3,0.79"

Prediction results:



KNN Prediction: Graduate
Naive Bayes Prediction: Dropout
Random Forest Prediction: Graduate

Figure 37: Result of the prediction

4. Conclusion

Through the analysis of student data such as academic achievement, attendance and socioeconomic characteristics, the system will enable institutions to focus their resources on the areas where it is the most needed. The focus of resources on these areas will ultimately lead to improved student retention.

During the development process of the solution, a dominance of the “Graduate” class was determined which may have influenced the model’s performance, particularly because of class imbalance. And based on the evaluation metrics, the predictions for the majority class “Graduate” were better compared to the minority class “Enrolled” across all of the models. This was likely caused by data imbalance in the dataset.

In terms of future work, many ways can be explored in order to solve this imbalance and improve the performance of the model. These techniques include-

- Resampling
- Assigning higher weight to minority class during model training.
- Using ensemble techniques such as bagging or boosting.

This system can also still be enhanced through the inclusion of additional features such as: mental health status, student engagement with extracurricular activities which can help improve predictions.

The machine learning model that makes use of Naive Bayes, KNN, and Random Forests offers a workable and efficient way to forecast student dropout. This has the potential to greatly increase retention rates, which would ultimately lead to better educational results by assisting educational institutions in early identification of at-risk students.

5. References

- aws, 2024. *What is Overfitting?*. [Online]
Available at: <https://aws.amazon.com/what-is/overfitting/>
[Accessed 25 December 2024].
- Chauhan, N. S., 2022. *Naïve Bayes Algorithm: Everything You Need to Know*. [Online]
Available at: <https://www.kdnuggets.com/2020/06/naive-bayes-algorithm-everything.html>
[Accessed 20 December 2024].
- Chi, Z., Zhang, S. & Shi, L., 2023. Analysis and Prediction of MOOC Learners' Dropout Behavior. *Applied Sciences*, Volume 13.
- Cote, C., 2024. *What Is Predictive Analytics? 5 Examples*. [Online]
Available at: <https://online.hbs.edu/blog/post/predictive-analytics>
[Accessed 20 December 2024].
- geeksforgeeks, 2024. *ML | Underfitting and Overfitting*. [Online]
Available at: <https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>
[Accessed 20 December 2024].
- geeksforgeeks, 2024. *Random Forest Algorithm in Machine Learnin*. [Online]
Available at: <https://www.geeksforgeeks.org/random-forest-algorithm-in-machine-learning/>
[Accessed 20 December 2024].
- geeksforgeeks, 2024. *What is Unsupervised Learning*. [Online]
Available at: <https://www.geeksforgeeks.org/unsupervised-learning/>
[Accessed 20 December 2024].
- Google Developer Program, 2024. *Classification: ROC and AUC*. [Online]
Available at: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
[Accessed 20 December 2024].
- Hayes, A., 2024. *Bayes' Theorem: What It Is, the Formula, and Examples*. [Online]
Available at: <https://www.investopedia.com/terms/b/bayes-theorem.asp>
[Accessed 20 December 2024].
- IBM, 2024. *What is a decision tree?*. [Online]
Available at: <https://www.ibm.com/think/topics/decision-trees>
[Accessed 20 December 2024].
- IBM, 2024. *What is predictive analytics?*. [Online]
Available at: <https://www.ibm.com/topics/predictive-analytics>
[Accessed 20 December 2024].

Kabathova, J. & Drlik, M., 2021. Towards Predicting Student's Dropout in University Courses Using Different Machine Learning Techniques.. *Applied Sciences*, Volume 11.

Park, H. & Yoo, S., 2021. Early Dropout Prediction in Online Learning of University using machine learning.. *International Journal on Informatics Visualization*, Volume 5.

Realinho, V., Machado, J., Baptista, L. & Martins, M. V., 2022. Predicting Student Dropout and Academic Success. *Data*, Volume 7.

Riebesell, J., 2024. *Random Forest*. [Online]
Available at: <https://tikz.net/random-forest/>
[Accessed 20 December 2024].

Simplilearn, 2024. *Classification vs. Clustering: Key Differences Explained*. [Online]
Available at: <https://www.simplilearn.com/tutorials/data-analytics-tutorial/classification-vs-clustering>
[Accessed 20 December 2024].

Tinto, V., 1975. Dropout from Higher Education: A Theoretical Synthesis of Recent Research. *Review of Educational Research*, 45(1).

Winograd, G., 2024. *College Dropout Rates 2025 (Statistics & Facts)*. [Online]
Available at: <https://missiongraduatenm.org/college-dropout-statistics/>
[Accessed 20 December 2024].